

CRM BANK PROJECT



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Problem statement

- Utilize customer demographics, transaction details, exit information, and active profiles to reduce churn, enhance service delivery, and improve customer satisfaction.
- Approach: 1. Analyze datasets to identify churn predictors and customer segments.
- 2. Implement visual to identify the churn and personalize interactions.
- 3. Measure success through churn reduction and different metrics





- ▶ **RowNumber:** The row number in the dataset, likely used for reference or indexing.
- ▶ **CustomerId:** A unique identifier for each customer.
- ▶ **CreditScore:** A numerical representation of the customer's creditworthiness.
 - ◆ **Credit score:**
 - Excellent: 800–850
 - Very Good: 740–799
 - Good: 670–739
 - Fair: 580–669
 - Poor: 300–579
- ▶ **GeographyID:** A numerical identifier that likely corresponds to a geographical location, such as a country or region.
- ▶ **GenderID:** A numerical identifier for the customer's gender, where for example, '1' could represent male and '2' could represent female.

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- Age: The age of the customer.
 - Tenure: The number of years the customer has been with the bank.
 - Balance: Current balance in the customer's account.
 - NumOfProducts: refers to the number of products that a customer has purchased through the bank.
 - HasCrCard: denotes whether or not a customer has a credit card. This column is also relevant, since people with a credit card are less likely to leave the bank.
 - 1 represents credit card holder
 - 0 represents non credit card holder
 - IsActiveMember: active customers are less likely to leave the bank (as per the criteria defined by the bank for identifying the activeness).
 - 1 represents Active Member
 - 0 represents Inactive Member
 - Estimated Salary: as with balance, people with lower salaries are more likely to leave the bank compared to those with higher salaries.
 - Exited: whether or not the customer left the bank.
 - 0 represents Retain
 - 1 represents Exit

Customer Churn and its impact on business

Customer churn, the rate at which customers stop using a company's products or services, is a crucial metric for banks.

It directly impacts revenue and profitability.

In this presentation, we will analyze our bank's customer churn rates, focusing on gender, recent years, customers with credit cards, number of products used, credit score-wise churn count, and geography-wise churn count.

Our goal is to identify factors contributing to churn and propose strategies to improve customer retention and satisfaction



DATA OVERVIEW

The dataset comprises seven tables including "Customer Info," "Churn," and "Active Customer," offering comprehensive insights into customer demographics and banking metrics.

"Customer Info" contains vital details like customer IDs, age, and estimated salary, forming the foundation for analyzing customer behavior and churn patterns.

Metrics such as account balances, tenure, and product holdings provide insights into customer engagement and financial behavior, aiding in churn prediction.

Predominantly categorical variables enable segmentation by geographic regions, gender, and credit score brackets, while continuous variables offer deeper analysis potential.



OBJECTIVE ANALYSIS



Account Balance Distribution

GeographyID	geographylocation	Total_balance	Average_balance	Max_balance	MIN_balance
1	France	311332479.49	62092.64	238387.56	0
2	Spain	153123552.01	61818.15	250898.09	0
3	Germany	300402861.38	119730.12	214346.96	27288.43

The SQL query computes the distribution of account balances across different regions by joining table's customer info, geography, and churn. It calculates the total balance, average balance, maximum balance, and minimum balance for each region. The results reveal the distribution of account balances for France, Spain, and Germany, showcasing their total balances, average balances, maximum balances, and minimum balances.

Churn rate by Gender

male

15.37

- The churn rate for females is 25.0488%, whereas the churn rate for males is 15.3716%.
- Overall, the churn rate for females is higher than the churn rate for males in the dataset you provided.



Female

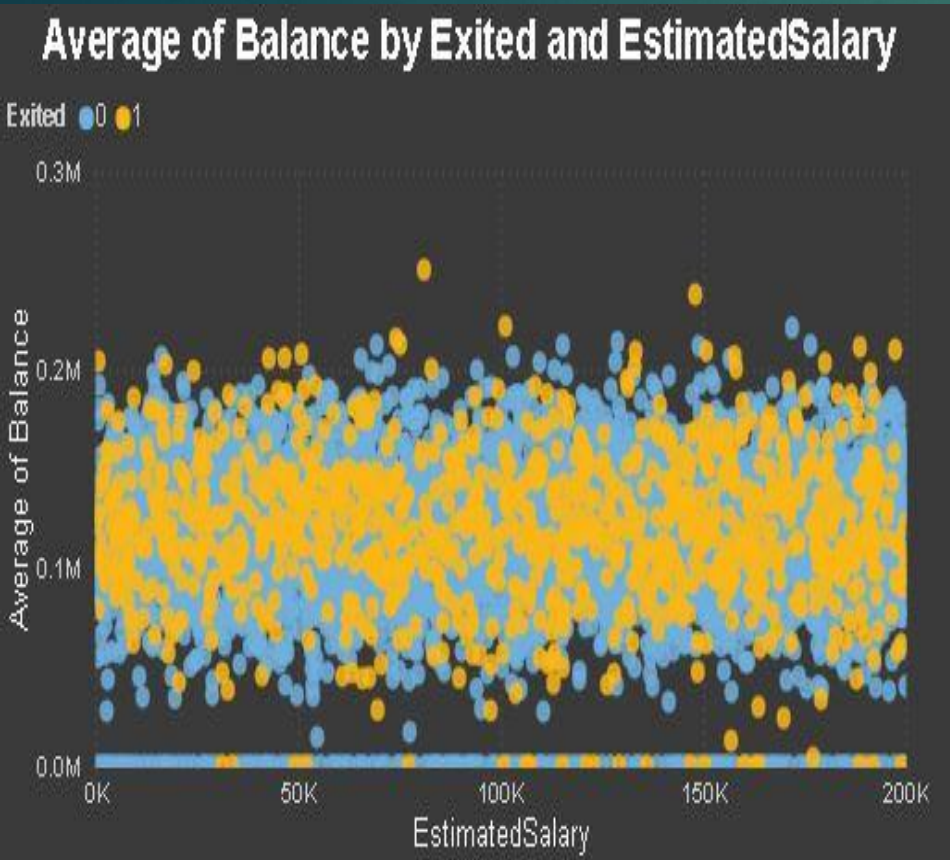
25.05

Gender-wise Income and Active Accounts:



Females have a higher average estimated salary (\$100,601.54) than males (\$99,664.58) in this. However, the table also shows that there are more active accounts for males (2,867) than females (2,284).

Correlation Between salary and balance



- There appears to be a positive correlation between the average balance and the estimated salary. This means that customers with a higher estimated salary tend to have a higher average balance.
- The data points are scattered around a general upward trend. This suggests that there is a relationship, but it is not a perfect one. There are many customers who deviate from the trend.

Subjective Insight's

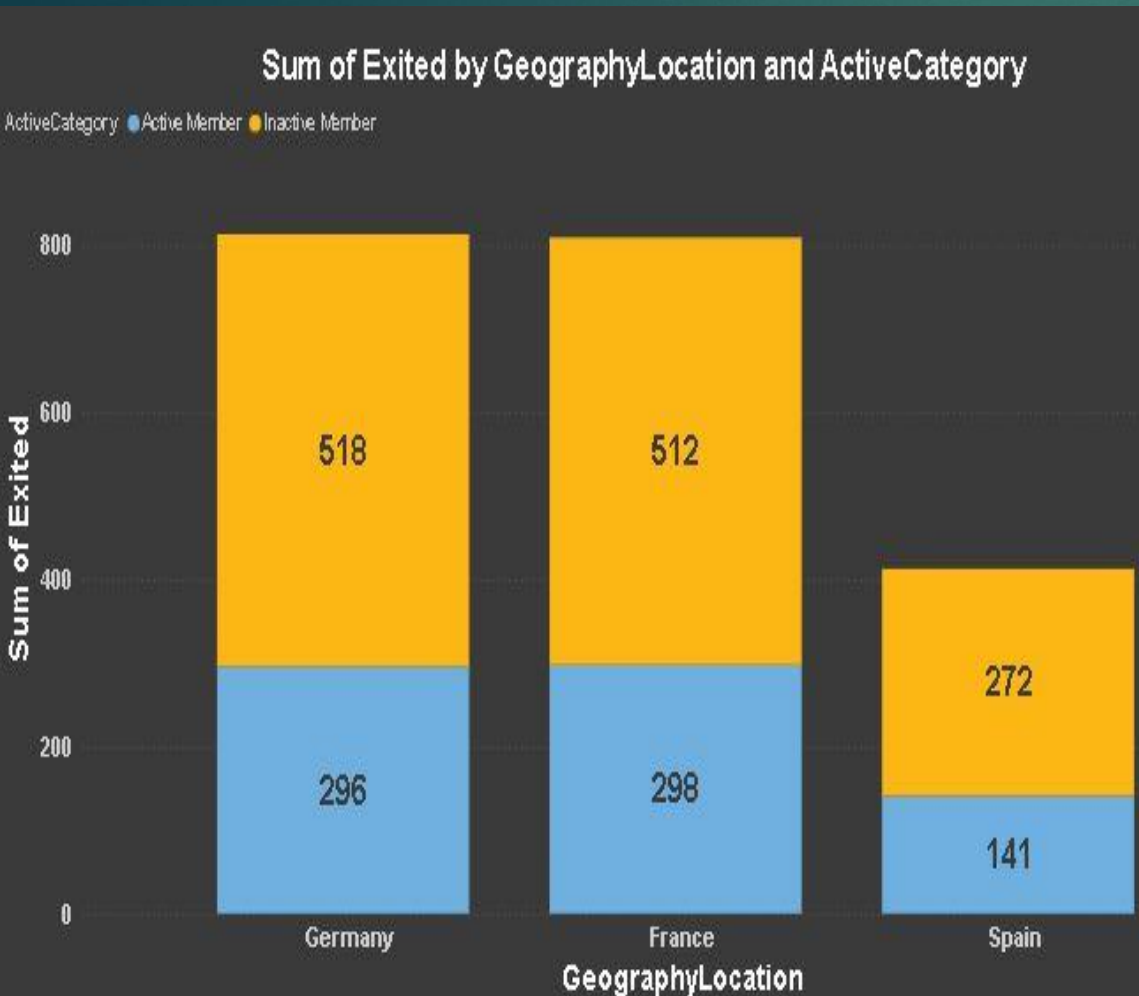
A man with a beard and two young boys are standing in a grassy field at dusk. The man is pointing towards a large blackboard in the background. The boy in the foreground is looking at the blackboard. The scene is dimly lit, with some lights visible in the distance.

Subjective

Poem
Analysis

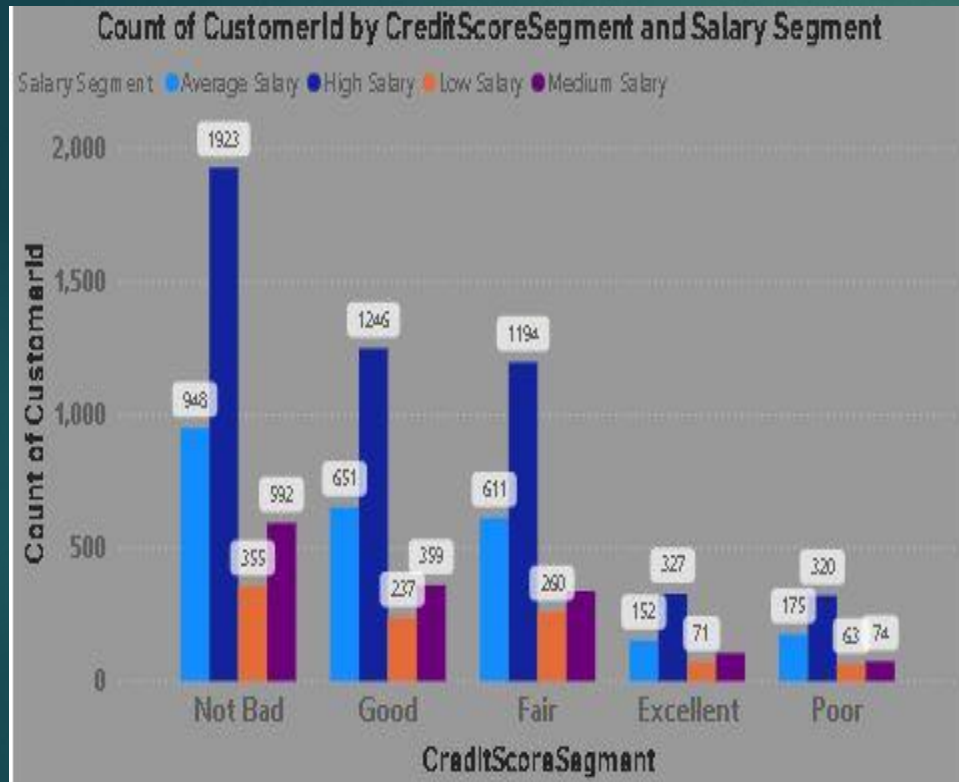
Subjective Insight's

Economic indicators in different geographic regions



There are more exited inactive members than exited active members in all three geographic locations. This suggests that inactive users are more likely to churn (stop using the product or service) than active users. This could be for a number of reasons,

Risk Management Assessment



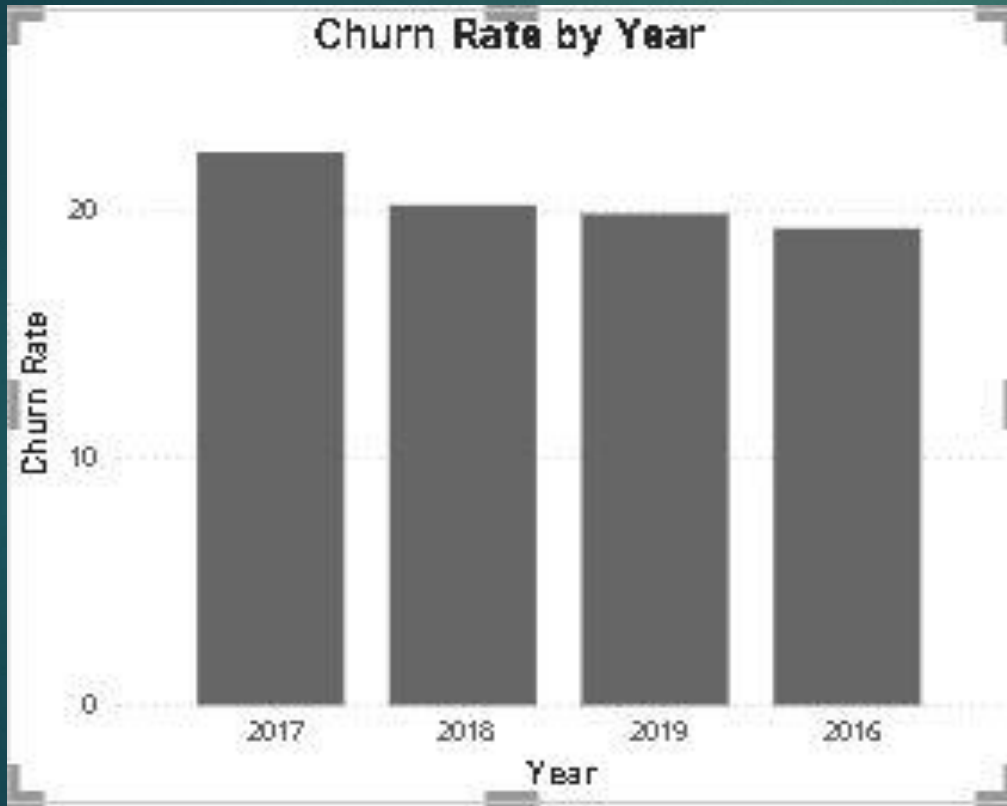
- Salary Segment: This column shows the salary brackets for the customers. The brackets include Low Salary, Medium Salary, High Salary, and Not Disclosed.
- Credit Score Segment: This column shows the credit score brackets for the customers. The brackets include Poor, Fair, Good, and Excellent.
- Count of CustomerID: This table shows the count of the number of customers in each combination of salary and credit score bracket. For instance, there are 237 customers with a Low Salary and a Poor credit score.

Customer Tenure Value Forecast



- High-value segments: Customers with high salary and high balance tend to have the longest tenure (4.88 years on average). This suggests that this segment may be the most valuable to the bank.
- Tenure by wealth: Looking at just the rich category, high balance customers still have a higher tenure than low balance customers (4.97 years vs 4.88 years). The same pattern holds for middle class and poor classifications.
- Tenure by salary: Within each balance category (high, medium, low) customers with high salary have a longer tenure than those with lower salaries.

Churn rate per year



Churn Rate:

- The overall churn rate for the bank is 20.37%.
- Year-on-year churn rates show some fluctuations:
 - 2016: 19.27%
 - 2017: 22.35% (highest)
 - 2018: 20.21%
 - 2019: 19.86% (lowest)

Conclusion

churn Rates by Gender:

- More females are leaving the bank (25.05%) compared to males (15.37%). This suggests that female customers may need more support or better services.

Salary and Activity:

- Females earn slightly more on average (\$100,601.54) than males (\$99,664.58), but there are more active male customers (2,867) than females (2,284). This could mean that while females have higher salaries, they are less engaged.

Salary and Balance Connection:

- Customers with higher salaries tend to have higher account balances. This shows that salary impacts how much money customers keep in their accounts, though not all high earners have high balances.

Inactive Customers at Risk:

- More inactive customers have left the bank than active ones across all areas. This means that keeping customers active is important for reducing churn.

Value of High Earners:

- Customers who earn well and have high balances stay with the bank longer (average of 4.88 years). This group is important for the bank's success.

Tenure and Salary:

- In every balance group, customers with higher salaries tend to stay longer. This indicates that higher salary customers are more loyal.

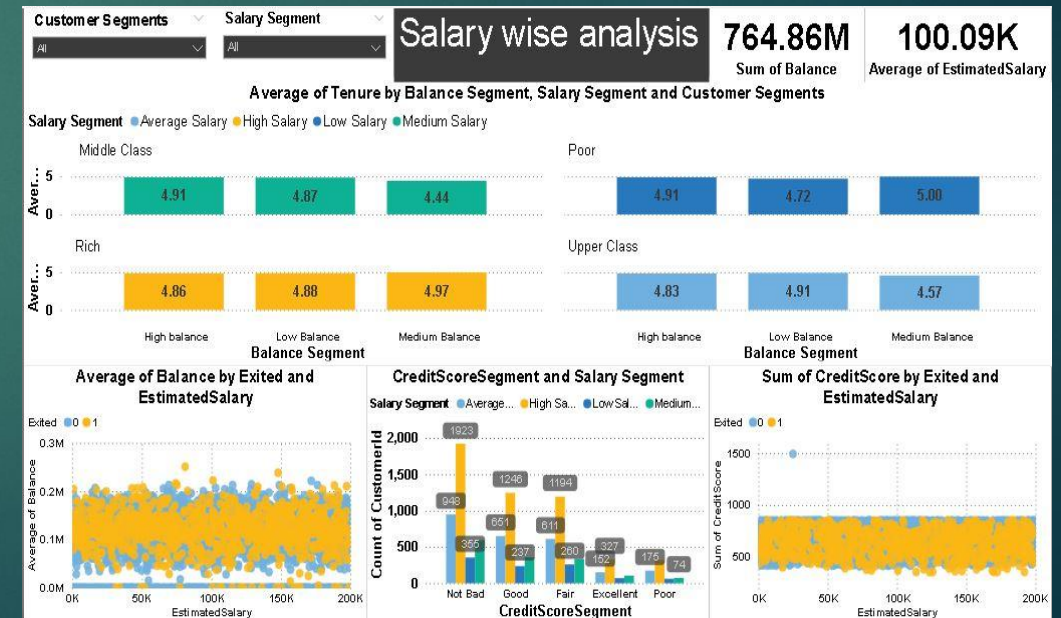
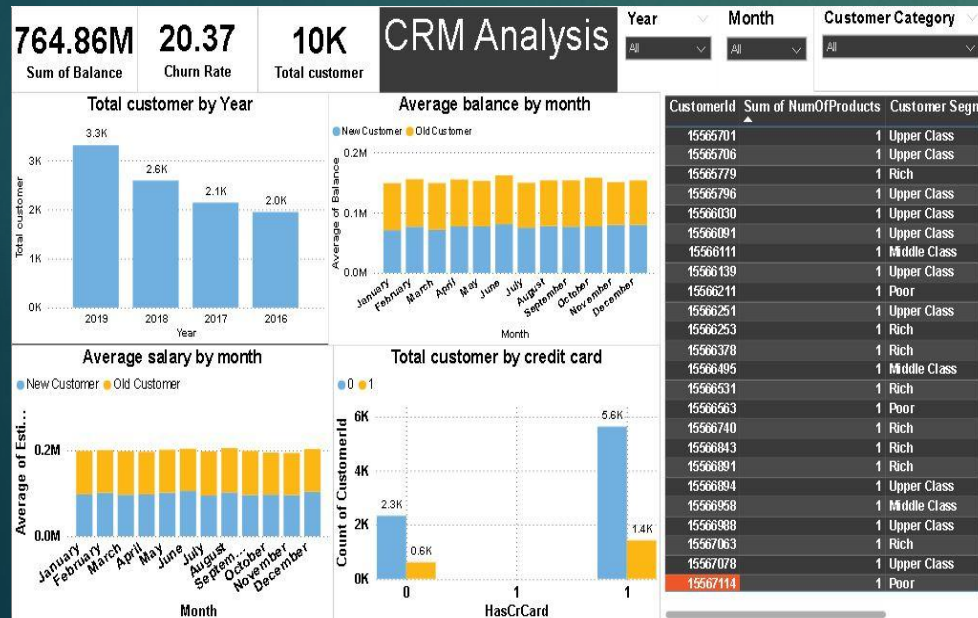
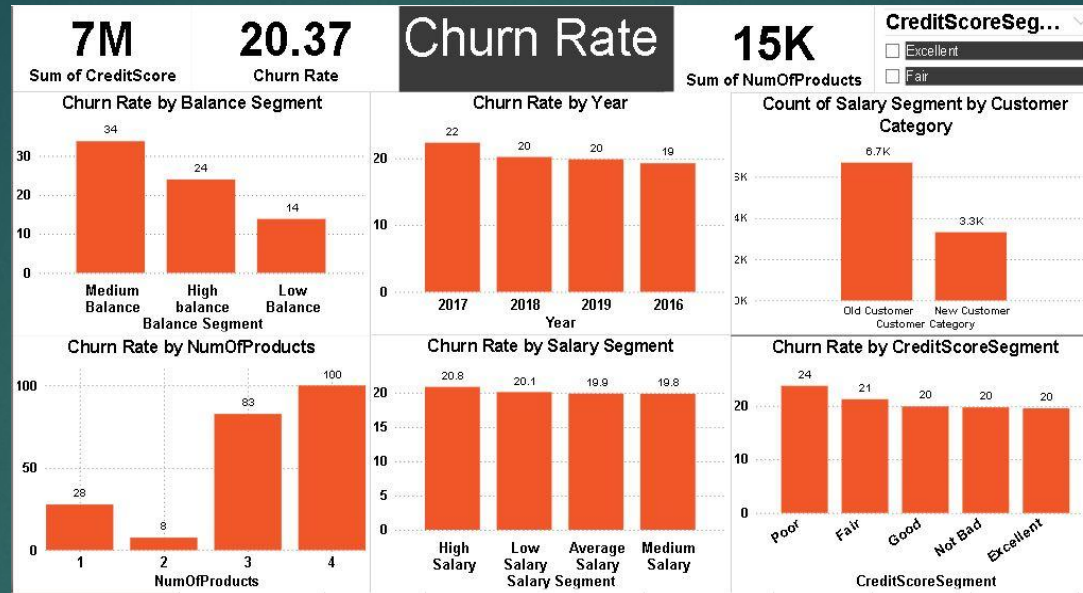
Churn Rate Over the Years:

- The overall churn rate is 20.37%, with the highest rate in 2017 (22.35%) and the lowest in 2019 (19.86%). This shows that customer retention fluctuates and needs ongoing attention.

Recommendations

1. **Support for Women:** Create special services and support just for female customers to make them feel valued and more engaged.
2. **Encourage Inactive Customers:** Launch programs to get inactive customers to use their accounts more, like offering promotions or sending personalized messages.
3. **Focus on High Earners:** Pay extra attention to customers with high salaries and balances, and offer them special services to keep them loyal.
4. **Track Churn Rates:** Regularly check how many customers are leaving so you can quickly spot and fix any problems.
5. **Personalize Services:** Use customer information to customize services and products to meet their individual needs, which will help improve satisfaction and reduce churn.

Dashboard





Thank You